

Reasoning

STATEMENT & CONCLUSIONS

In almost all the Objective-Multiple Choice type examinations whether it is Bank Probationary Officers, Bank Clerks, Staff Selection Commission Combined Graduate Level Tier-I Exam, SSC SAS Exam, SSC CISF ASI Exam, SSC Higher Secondary Level Stenographer (Grade 'C and Grade 'D') Exam, SSC Higher Secondary Level Data Entry Operator and Lower Division Clerk Exam, Management Entrance Tests, Navy, Air Force and Insurance Sector Exam, some questions are asked from Statement and Conclusions. In this type of questions some statements, containing certain pieces of information, are given followed by two or more inferential conclusion which can be derived from the given pieces of information in the statement (s). You are required to consider both the statement(s) and conclusions and then decide which of the conclusions logically follows beyond a reasonable doubt. Here, reasonable doubt implies that the conclusion must be supported by the pieces of information given in the question statement. There is no scope for any assumption and imagination. The derived conclusion should not raise any doubt. In other words, the validity of conclusion should not be questioned on any ground. In no circumstances the validity of the conclusion should be doubted. At this stage of discussion consider the following example :

Ex. A is in the room when B is in the Guest House. Here one question may arise.

Can A in the room when B is not in the Guest House?

The answer to this question will be affirmative, without any reasonable doubt. The condition given in the first statement is that is B is in the Guest House, then A definitely has to be in the room. A can be in the room otherwise also even when B is not in the Guest House.

The valid conclusions you can make about the principal statement given would be :

(a) B is in the Guest House. A is in the room.

(Straight or Direct Logic)

(b) A is not in the room, B is not in the Guest House.

(Straight Negation)

Conclusion (b) is also correct because at any given point of time, A and B have two possible positions each, i.e., A can be in the room or outside.

B can be in the Guest House or outside.

When A is in the room, B has two options, of either being in the Guest House or outside.

The questions on **Statement and Conclusion** are meant to test your ability to derive correct and valid conclusions from the pieces of information given in the statement (s). The most important aspect of such questions is the nature of conclusions. Some of the conclusions can be easily and quickly detected because

these follow directly from the facts stated in the question statement. The pieces of information given in the question statement reasonably supports such conclusions. But sometimes conclusions may be indirect. In such cases it becomes very difficult to decide the validity of the conclusion.

A conclusion will follow, if

(i) it is implied in one or all of the pieces of information given in the statement (s).

(ii) it is generalisation of the ideas contained in the statement (s).

(iii) no material fallacy is committed, viz.,

(a) fallacy of composition.

(b) fallacy of division.

(c) fallacy of uncritical analogy.

(d) fallacy of accident and

(e) fallacy of insufficient data.

How to attempt such type of questions ?

If more than one statement are given, the information given in different statements should be co-related to arrive at a conclusion.

Ex. 1. Statements : Most people are good at heart and non-manipulative.

A few who become manipulative have been found to be lacking in keeping strong family ties.

Which of the following can be concluded from the above statements ?

(1) People from different socio- economic background behave differently.

(2) Being manipulative is not considered as a virtue.

(3) Family plays the most important role in one's upbringing.

(4) Non-manipulative persons are not highly successful.

(5) People like to be manipulative.

[Bank P.O. Exam, 2008]

Explanation : If we co-relate the given information we will get:

Most people are good + some manipulative.

Again, manipulative persons cannot maintain strong family ties and hence such persons are not successful.

Therefore, our required answer would be option (4).

Ex.2. Statements : The books of Kiran Prakashan are well written.

These books are useful for one day competitive examinations.

Explanation : If the two pieces of information are co-related, we will get :

Well written books + useful for examination

Valid Conclusions:

I. The books of Kiran Prakashan are useful for objective type examinations

II. Well written books are useful.

- You should consider the validity of conclusion only in the light of information given in the statement.
- The Rule of identity" must be followed in each of the statements. In other words, a phrase or word should not be used in two different senses.

- If two statements are given, these should not be self-contradictory, otherwise no conclusion can be derived from them.

Ex. Statements : The average number of students per teacher is fifty in urban areas, whereas it is sixty in rural areas.

In urban areas, there is one teacher for every sixty students.

Explanation : The two statements are self-contradictory. We cannot derive definite conclusion from them.

- The statement (s) and the following conclusions must be factually correct. They should not go against the established fact or prevailing notion of truth.

Ex. Statement: The President of India is elected through secret ballot.

Conclusion : He is, thus not an elected head of the State.

Explanation : The conclusion is factually incorrect.

- From examples and precedents, we cannot derive any conclusion.

Ex. Statement: Gandhiji was a great personality who preached non-violence.

Invalid Conclusion : One who preaches non-violence is a great personality.

- From a very general statement, no definite conclusion can be drawn. Therefore, certain words such as **Generally, Usually, Normally** etc. when present in the statement, make conclusion invalid or doubtful.

Ex. Statements :

Morning walk is good for health.

People generally goes for morning walk!

Rashmi generally goes for morning walk.

Therefore,

Rashmi is healthy.

Explanation : The use of word "generally" in the second and the third statements makes the conclusion doubtful.

- Certain words when used in the statement (s) may alter, emphasise or substantiate the meaning of the statement. These words when present in the statement give important clues for detecting valid or invalid conclusions. Such words are : **All, Some, None, Never, Always, Sometimes, Must be, May be, Might be, Will be, Although, Despite, Only, Most of, Majority of etc.**

Ex.1. Statement: Most of the people convey that they get bright and excellent ideas in the early morning hours.

Conclusion : Ideas come only in the early morning hours.

[Bank P.O. Exam, 2009]

Explanation : "Most of the people" does not imply "all the people". Thus, some people could get bright and excellent ideas anytime. The use of word "only", thus, in the conclusion makes it invalid.

Ex.2. "All trees bear fruit". Fruits of some trees are not good for human health".

If the two statements are assumed to be true then which of the following statements is **definitely true?**

(1) All fruits are not sweet & tasty.

(2) Some fruits are healthy for some human beings.

(3) Some trees do not bear fruits.

(4) Human beings should eat fruits for remaining healthy.

(5) None of these

[Bank P.O. Exam, 2008]

Explanation : All trees bear fruit -> No exception

The second statement implies that fruits of some trees are good for human health.

Now analyse each conclusion one by one :

Conclusion (1) : All fruits are not sweet and tasty.

The use of word "all" makes it invalid.

Conclusion (2) : Why some fruits are healthy for only some people?

Conclusion (3) : It is given that "all trees bear fruit without exception and hence the conclusion is invalid.

Conclusion (4): Since fruits of some trees are good for health, we should eat fruit for remaining healthy.

- If a single statement is followed by more than one conclusion, then evaluate each conclusion separately to detect the valid and invalid conclusions.

Ex.1. Most of the people convey that they get bright and excellent ideas in the early morning hours.

(1) Ideas come only in the early morning hours.

(2) Chances of getting bright ideas at night are less.

(3) People are not able to express clearly how they get ideas.

(4) After sound sleep of two hours mind becomes fresh.

(5) Unless the ideas are bright, it does not serve any purpose.

Ex.2. Over the years the average age of both men and women in India has gone up considerably over the last 70 years.

(1) Rekha would probably live longer than her grandmother.

(2) There is a wider difference between the average age of men and women.

(3) There seems to be a small difference between the average age of men and women.

(4) Drugs to increase longevity have been found.

(5) The average age of marriage has also gone up over the years.

[Bank P.O. Exam, 2009]

- In some cases the conclusion clearly or directly follows from the information given in the statement, while in some others the conclusions may be re-statement of the given statement. It is very easy to judge such conclusions. Consider the following examples :

Ex.1. Statement : Only those candidates with exceptional talent and strong motivation should apply - An advertisement.

Conclusion:

Candidates not fulfilling these criteria will not be considered.

[Bank P.O. Exam, 2008]

Explanation : Conclusion logically follows from the statements. It is categorically mentioned that only talented and motivated candidates should apply. Therefore, candidates not fulfilling these criteria will not be considered.

Ex.2. Statement: Imports have led to a difficult foreign exchange position.

Valid Conclusions

I. Imports require foreign exchange.

II. Foreign exchange is spent for importing goods.

In the foregoing examples, we observe that it is very easy to evaluate the valid conclusions as they directly follow

from the statement. But sometimes the conclusions are not so direct and they require application of some logic on your part. You will have to make certain assumptions or apply your commonsense; and using these assumptions together with the information given in the statement (s), the given conclusions can be evaluated.

Now consider the following examples :

Ex.1. Statement : Only those candidates with exceptional talent and strong motivation should apply- An advertisement.

Conclusion : It is possible to decide clearly who is talented and motivated.

[Bank P.O. Exam, 2009]

Explanation : If we assume that the advertisement was issued only when it was assured of that it is possible to decide talented and motivated person, the conclusion is valid.

After giving details of various points pertaining to the questions on STATEMENT AND CONCLUSIONS now consider some questions asked in the previous Bank Probationary Officers' Exams.

EXAMPLES

Directions (1-15) : In each question below is given a statement followed by two conclusions numbered I and II. You have to assume everything in the statement to be true, then consider the two conclusions together and decide which of them logically follows beyond a reasonable doubt from the information given in the statement.

Given answer (1) if only conclusion I follows.

Given answer (2) if only conclusion II follows.

Given answer (3) if either I or II follows.

Given answer (4) if neither I nor II follows.

Given answer (5) if both I and II follows.

Ex.1. Statement : The oceans are a storehouse of practically every mineral, including uranium. But like most other minerals, it is found in extremely low concentration—about three gms. per 1000 tonnes of water.

Conclusions

I. The oceans are a cheap source of uranium.

II. The oceans harbour radiation hazards.

Ex.2. Statement : The Minister questioned the utility of the space research programme and suggested its replacement by other areas of felt national needs.

Conclusions

I. Exploring the space does not contribute to critical national needs.

II. Research should be oriented to national needs.

Ex.3. Statement : The laws and statutes framed by the Government for the purpose of providing equal treatment to every citizen, on implementation perpetuate corrupt working system.

Conclusions

I. The laws and statutes should be framed but they should not be implemented to avoid corrupt working system.

II. There should be obvious method to investigate corrupt working system.

Ex.4. Statement : Mrs. X is nominated for one of the two posts of which one post is reserved by the Managing Committee for the female of other religious minority community and the other for the female of Scheduled Castes or Scheduled Tribes.

Conclusions

I. Mrs. X is the member of religious minority community.

II. Mrs. X is the member of Scheduled Castes or Scheduled Tribes.

Ex.5. Statement : We do not need today in India extraordinary specialists but those trained ordinary doctors who are dedicated to their profession.

Conclusions

I. We should promote medical profession with dedicated ordinary doctors rather than promoting high specialised medical education.

II. Extraordinary specialists are not dedicated to their profession.

Ex.6. Statement : Although we have rating agencies like Crisil, ICRA, there is demand to have a separate rating agency for IT Companies to protect investors.

Conclusions:

I. Assessment of financial worth of IT Companies calls for separate set of skills, insight and competencies.

II. Now the investors investing in IT Companies will get protection of their investment.

Ex.7. Statement : Company "Y" will improve the manufacturing facilities for the production of shaving kits as a result of which capacity would increase and cost would be reduced—A spokesperson of the Company "Y".

Conclusions :

I. The products of Company "Y" will complete the market norms in the quality and cost factor.

II. There will be demand of shaving kits of Company "Y".

Ex.8. Statement : During 1997-98 the total loss incurred by the 111 Public Sector Units was to the tune of Rs. 6809 crore which was converted into paid capitals by the Government of its total investment of Rs. 5129 crore.

Conclusions :

I. The Government is left with only one option that is to privatise these units.

II. The Government did not take care in the matter of investments in these public sector units.

Ex.9. Statement : Population increase coupled with depleting resources is going to be the scenario of many developing countries in days to come.

Conclusions:

I. The population of developing countries will not continue to increase in future.

II. It will be very difficult for the governments of developing countries to provide its people decent quality of life.

Ex.10. Statement : Mr. X is one of the probable candidates short-listed for the post of Director of K.L.M. Institute.

Conclusions:

I. Mr. X will be selected as Director of K.L.M. Institute.

II. Mr. X will not be selected as Director of K.L.M. Institute.

Ex.11. Statement: The cabinet of State 'X' took certain steps to tackle the milk glut in the state as the cooperatives and Government dairies failed to use the available milk—A news report.

Conclusions :

I. The milk production of State 'X' is more than its need.

II The Government and co-operative dairies **in** State '**X**' are not equipped in terms of resources and technology to **handle** such excess milk

Ex.12. Statement : My first and foremost task is to beautify this city— if City 'X' and 'Y' can do it- why can't we do it— Statement of Municipal Commissioner of City 'X' after taking over charge.

Conclusions :

I. **The** people of City 'X' are not aware about the present state of ugliness of their city.

II. The present Commissioner has worked in City '**X**' and 'Y' and has good experience of beautifying cities.

Ex. 13. Statement: Women's Organisations in India have welcomed the amendment of the Industrial Employment Rules 1946 to curb sexual harassment at the work place.

Conclusions:

I. Sexual harassment of women at work place is more prevalent in India as compared to other developed countries.

II. Many organisations in India will stop recruiting women to avoid such problems.

Ex.14. Statement : It has been decided by the Government to withdraw 33% of the subsidy on cooking gas from the beginning of next month— A spokesman of the Government.

Conclusions:

I. People now no more desire or need such subsidy from Government as they can afford increased price of the cooking gas.

II. The price of the cooking gas will increase at least by 33% from the next month.

Ex.15. Statement : 'The Government will review the present policy of the diesel price in view of further spurt in the international oil prices'— A spokesman of the Government.

Conclusions:

I. The Government will increase the price of the diesel after the imminent spurt in the international oil prices.

II. The Government will not increase the price of the diesel even after the imminent Spurt in the international oil prices.

EXPLANATIONS

1.(4) Neither of the conclusions follows. It has been stated that uranium is found in very small quantity, i.e., about three grams per 1000 tonnes of water. Therefore, ocean cannot be a cheap source of uranium.

2.(5) Both the conclusions follow beyond a reasonable doubt.

3.(2) If laws **and statutes** framed are not implemented then what is the need of framing laws and statutes. But conclusion II definitely follows.

4.(3) It is clear that Mrs. X is a member of either religious minority community or Scheduled Castes/Scheduled Tribes.

5.(1) From the statement it is clear that conclusion I follows beyond a reasonable doubt.

6.(1) Only conclusion I seems to be reasonable. Considering the different nature of IT Companies different **parameters** should be employed for ratings. **It** is not necessary that if separate rating agency is established for IT Companies the investors will get protection of their investment. Therefore, conclusion II does not follow.

7.(1) By increasing the manufacturing capacity the Company 'Y' would compete reasonably on the cost front. And, as such it can improve the quality of its products. Hence, conclusion I follows. Conclusion II seems to be an assumption.

8.(2) Considering the amount of loss incurred by Public Sector Units it seems to be true that the Government did not take care in the matter of investment in the Public Sector Units. The use of term "only" in the conclusion I makes it invalid.

9.(2) Clearly, only conclusion II follows. It is not clear how the population of developing countries will not increase in the future.

10.(3) It is mentioned in the statement that Mr. X has been declared successful in the preliminary screening for the post of Director of KLM Institute. Therefore, either he will be selected or will not be selected as Director of KLM Institute.

11.(5) Both the conclusions logically follow from the information given in the statement. The term **milk glut** clearly indicates that the milk production of State 'X' is more than its requirement. Again, it is clearly mentioned that the Government and co-operative dairies in State 'X' failed to use the available milk—it implies that such dairies in State 'X' are not equipped properly.

12.(4) Neither of the conclusions follows. It is clear from the statement that the City 'Z' requires beautification but this does not imply that the people of that city are unaware about the state of ugliness of their city. Therefore, conclusion I does not follow. The new Municipal Commissioner asserted that the task of beautification of City 'Z' could also be accomplished if such has happened in City 'X' and City 'Y'. From this it cannot be derived that the Municipal Commissioner has served in City 'X' and City 'Y'. While asserting something one may quote the words of another person. Hence, conclusion II is also not valid.

13.(4) Neither conclusion I nor conclusion II follows. In the absence of harsh punitive measures the cases of sexual harassment may take place anywhere. Therefore, it is erroneous to assume that sexual harassment of women at work place is more prevalent in India as compared to other developed countries. The statement exhorts that amendment in the Act is meant to curb sexual harassment at the work place and it cannot be taken as the imposition of restriction on recruitment of women. In order to avoid pain in arms we cannot chop our arms.

14.(4) Neither conclusion I nor conclusion II follows. The subsidy on any item cannot be 100 per cent. It is mentioned in the statement that the Government has decided to withdraw 33 per cent of "the subsidy" on cooking gas and not 33 per cent of the actual price of the cooking gas. Therefore, it is erroneous to assume that the price of the cooking gas will increase **at least by 33 per cent**. Any policy of the Government affects all sections of the society and it cannot be assumed that poor people do not need subsidy. Thus, neither of the conclusions follows.

15.(3) From the statement it is clear that the Government will either increase the price of diesel or will not increase the price of diesel. The deficit on this count can be adjusted by some other means. Therefore, after the review of the present policy of the diesel price in view of further spurt in the international oil prices, the Government is left with only two options—to increase or not to increase the price of diesel. Therefore, either conclusion I or conclusion II follows.